THE FIRST KOSZUL HOMOLOGY GROUP IS ANNIHILATED BY THE GENERATING IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The group H_1(N, f_bullet) is defined as cycles of the form (x_1, ..., x_n) with f_i x_j = f_j x_i modulo Koszul boundaries, per the explicit Ext lemma. invoked at statement lines 24455-24458. Fixes the cycle and boundary description the computation acts on.

Source reference. Original lemma 05EE: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24413–24413. Cited at statement lines 24455-24456 for the definition of H_1(N, f_bullet) the computation acts on; the description is load-bearing inside the counted span.

Lemma 0.2. *Let R be a ring. Let $I = (f_1, \ldots, f_n)$ be a finitely generated ideal of R. For any R-module N the Koszul homology group $H_1(N, f_\bullet)$ defined in Lemma 05EE is annihilated by I.*

Proof. Let $(x_1, \ldots, x_n) \in N^{\oplus n}$ with $f_i x_j = f_j x_i$. Then we have $f_i(x_1, \ldots, x_n) = (f_i x_i, \ldots, f_i x_n)$. In other words f_i annihilates $H_1(N, f_\bullet)$. □

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